

# Rex Ouyang

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## EDUCATION

### University of California, Berkeley

May. 2028

B.A. Computer Science, B.A. Applied Mathematics; EECS Honors Program

Berkeley, CA

- **Coursework:** Nonstochastic Statistics\*, Stochastic Estimation and Control\*, Applications of Stochastic Process Theory\*, Random Processes in Systems\*, Convex Optimization\*, Topology and Analysis\*, Deep Learning

\* denotes graduate coursework

## SELECTED PROJECTS & EXPERIENCE

### Undergraduate Student Researcher, UC Berkeley EECS

May. 2026 – Present

Researching sharp entropy bounds in Euclidean spaces and geometricity conditions under Prof. Thomas Courtade

- Studying information-theoretic functional inequalities using Convex Analysis and Stochastic Calculus
- Analyzing reductions from entropy optimization over probability measures to finite-dimensional covariances

### Undergraduate Course Staff, UC Berkeley EECS

Aug. 2026 – Present

1 out of 3 Selected Tutors for UC Berkeley EECS126: Probability and Random Processes, for Fall 2026 semester

- Will lead office hours, answer student questions, grade assignments and exams, and support administration

### Reinforcement Learning Environment for Deterministic Platformer Control

Aug. 2026 – Present

Designing an AlphaZero-style planning agent for playing Geometry Dash, exploiting deterministic fixed-tick physics to combine a learned policy with real-time forward search

- Building the state representation from direct game-engine data rather than rendered frames for CNNs
- Implementing state forking against the C++ engine to roll candidate action sequences forward N ticks, enabling MCTS to plan through obstacle layouts instead of memorizing level-specific trajectories

### An Exposition of the Sharp Symmetrized Talagrand Inequality

May. 2026 – Aug. 2026

Research project studying a [recent result](#) on optimal transport bounds measuring distance to centered distributions

- Using Convex Analysis and Stochastic Calculus, provided alternate proofs to key results in [this report](#)
- Built Monte Carlo simulations on probability distributions to test and visualize the inequality numerically

### Vectorized Entropy-Based Wordle Solver

Mar. 2025 – Jul. 2026

Built an information-theoretic solver for NYT Wordle that treats each game as sequential Bayesian inference, choosing the guess that maximizes expected entropy gain over the posterior of remaining candidate words

- Designed a two-tier candidate model that falls back from the 2,315 curated answers to the full 12,972-word legal list, making the solver robust to out-of-vocabulary answers that break naive filtering
- Averaged 3.483 guesses per game across simulated play, outperforming the human average of 3.92 guesses

### Simulations Developer, Formula Electric at Berkeley

Sep. 2024 – May. 2026

Analyzed measured current-draw time-series to model battery discharge dynamics under varying load, deriving a reduced formulation that estimated battery state faster than the team's existing numerical simulation

- Contributed the model to the shared simulation codebase for integration in a multidisciplinary team

### Sparse Attack Detection in Power Networks via Lasso Optimization

Nov. 2025 – Dec. 2025

Formulated sensor attack detection as a sparse recovery problem as an L1-regularized regression problem over measurements corrupted by gaussian noise and sparse adversarial injections.

- Built a Monte Carlo experiment generating corrupted measurement datasets across 100 randomized trials
- Achieved exact support recovery, every attacked node and line with zero false positives, in 95% of trials

## TECHNICAL SKILLS

- **Quantitative Methods:** Probability, Statistics, Stochastic Processes, Stochastic Analysis, Convex Optimization, Sparse Recovery, Statistical Modeling, Information Theory, Dynamical Systems, Machine Learning
- **Programming / Systems:** Java, Python, C/C++, SQL, Go, RISC-V Assembly, x86 Assembly
- **Scientific Computing / Tools:** NumPy, SciPy, CVXPY, Matplotlib, Git, Claude CLI, Codex CLI